



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
FULL SYLLABUS
28-08-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

PAPER-2

Time : 2:45 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 18)

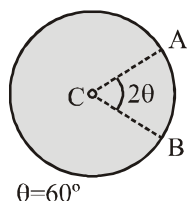
- This section contains **SIX** questions.
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Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : -1 In all other cases

1. A rocket is launched from point A on earth. It reaches point B on earth with its velocity antiparallel to its initial velocity. The angle subtended by line AB at the center is 2θ . The satellite moving close to surface of earth has time period of revolution T_0 . Radius of earth is R. Ratio of time taken by the rocket from A to B to the time period of revolution of satellite is :-



(A) 1/3

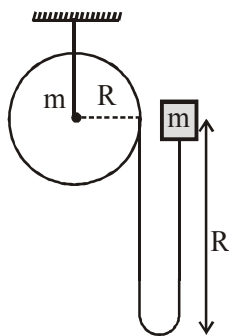
(B) 2/3

(C) 1/2

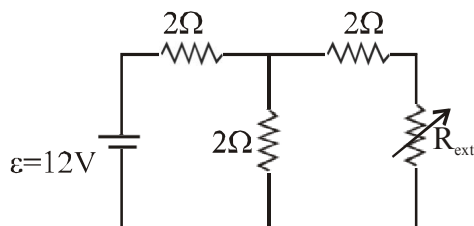
(D) 3/4

Space for Rough Work

2. Mass of the pulley is m & radius is R . Assume pulley to be disc. Block of mass m is released from the position shown. String is massless & inextensible. There is no slipping between rope and pulley. The impulse exerted by the string on the pulley at the moment string becomes taut is J . $\frac{2m\sqrt{gR}}{J}$ is equal to.



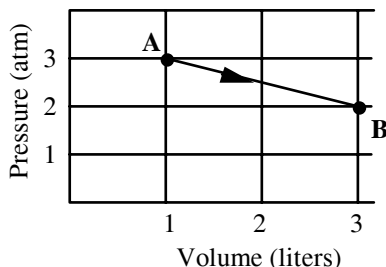
- (A) 3 (B) 4 (C) 5 (D) 6
3. Value of maximum power (in Watt) delivered to R_{ext} is



- (A) 3 (B) 6 (C) 9 (D) 12

Space for Rough Work

4. An ideal monatomic gas originally in state **A** is taken reversibly to state **B** along the straight-line path shown in the pressure-volume graph. The change in the internal energy of the gas for this process is X Joule. Value of $\frac{X}{50}$ is.



- (A) 5 (B) 4 (C) 9 (D) 15
5. A “soft x-ray” photon with an energy of 41.8 eV is absorbed by a hydrogen atom in its ground state, knocking out the atom's electron. The speed of the electron as it leaves the atom is $p \times 10^6$ m/s. The value of p is
- (A) 1.84×10^6 m/s (B) 3.08×10^6 m/s (C) 8.16×10^6 m/s (D) 3.15×10^6 m/s
6. The half-life of a substance A is 1 hr. The half-life of another substance B is 10 years. A mixture of the two substances has an activity of 1000 Bq. After 1 hr, its activity drops to 900 Bq. What will be its activity after 20 years ?
- (A) 10^2 Bq. (B) 5×10^2 Bq. (C) 3×10^2 Bq. (D) 2×10^2 Bq.

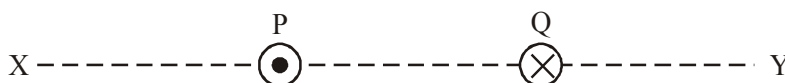
Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

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 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
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-
7. A boy is standing on a platform which is accelerating upwards with an acceleration 2m/s^2 . Initially the platform was at rest on the ground. At $t = 2$ sec, the boy throws a ball upwards with a velocity 8 m/s relative to himself. Choose the correct alternatives:
- (A) he will catch the ball $4/3$ sec after he throws the ball.
(B) ball attain a maximum height of $8/3\text{ m}$ w.r.t. platform.
(C) if he does not catch the ball, then ball strikes the ground after 4 sec.
(D) none of the above is correct.

Space for Rough Work

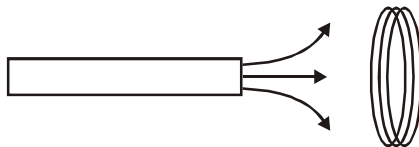
8. A light rod of 2m length is suspended from the ceiling horizontally using two vertical wires of equal length tied to its ends. One of the wire is made of steel of cross-sectional area 0.1 sq. cm and Young's modulus 20×10^{11} dyne/cm², while the second is made of brass of cross-sectional area 0.2 sq. cm. and Young's modulus 10×10^{11} , then choose the correct alternatives
- (A) For stress to be same, a mass can hang at $2/3$ m from brass wire end.
 (B) For stress to be same, a mass can hang at $2/3$ m from steel wire end.
 (C) For equal strain, the mass should be hung at mid point of the rod.
 (D) For equal force, the mass should be at $1/4^{\text{th}}$ from steel end
9. When a gas is heated in a vessel of constant volume, the pressure increases because
- (A) The average force per impact of a molecule on a wall of the vessel increases
 (B) Molecules hit the walls more frequently
 (C) Molecules collide with each other more frequently
 (D) The number of molecules per unit volume of the gas increases
10. Two long and parallel wires at P and Q carry currents 0.1 A and 0.2 A respectively in opposite directions as shown. In which region(s) magnetic field is not zero?



- (A) Between X and P
 (B) Between P and Q
 (C) Between Q and Y
 (D) In all the regions there exists no neutral point.

Space for Rough Work

11. You are given a bar magnet and a coil of wire. Which of the following would induce an emf in the coil ?



- (A) Moving the magnet toward the coil. (B) Moving the coil away from the magnet.
 (C) Turning the coil about a vertical axis. (D) none of these
12. The surface of a spherical star pulsates. This pulsation can be assumed to be the fundamental mode of vibration with antinode at the surface of the star and node at the centre. If the amplitude of surface is A , bulk modulus of the material is B and density is ρ , choose the correct statement(s)

(A) The frequency of pulsation is $\frac{1}{4R} \sqrt{\frac{B}{\rho}}$

(B) The amplitude at a distance $\frac{R}{2}$ from the centre is $\frac{A}{2}$

(C) The amplitude at a distance $\frac{R}{2}$ from the centre is $\frac{A}{\sqrt{2}}$

(D) The maximum velocity of the surface of the star is $\frac{\pi A}{R} \sqrt{\frac{B}{\rho}}$

Space for Rough Work

13. A time varying voltage $V = 10 - 2i$ (volt) is applied across an inductor of inductance $L = 2H$ where i is current at any time t with $i = 0$ at $t = 0$. Then
- (A) current increases with time
 - (B) energy stored in magnetic field at $t = \ln 2$ sec is $6.25 J$
 - (C) at time $t = \ln 2$ sec energy stored in magnetic field is increasing at a rate of $12.5 J/sec$
 - (D) energy stored in magnetic field is zero all the time
14. A particle of mass m and charge q is projected in a region where an electric field given by $\vec{E} = E_0 \hat{i}$ exists with a velocity $v_0 \hat{j}$ from the origin at time $t = 0$, then choose the correct statements (assuming $m^2 v_0^2 = 2qE_0 m x_0$).
- (A) radius of curvature of the particle when its x-coordinate becomes x_0 is $2x_0$.
 - (B) radius of curvature of the particle when its x-coordinate becomes x_0 is $4\sqrt{2}x_0$.
 - (C) speed of the particle when its x-coordinate becomes x_0 is $\sqrt{2}v_0$.
 - (D) speed of the particle when its x-coordinate becomes x_0 is $2v_0$.

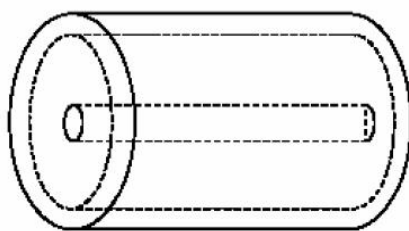
Space for Rough Work

SECTION-I(iii) : (Maximum Marks : 12)

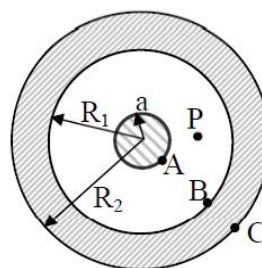
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Paragraph for Questions 15 and 16

A very long cylindrical plastic rod of radius $a = 2.5$ cm has electric charge per unit length of density $\lambda = +6.0 \times 10^{-7}$ C/m distributed uniformly throughout its volume. This rod is surrounded by a concentric cylindrical metal shell of inner radius $R_1 = 7.5$ cm and outer radius $R_2 = 10$ cm, as shown. The metal shell is initially uncharged.



(side view)



(front view)

15. The potential difference between point A on the rod and point C on the outer shell is nearly
 (A) 12 kV (B) 6 kV (C) 24 kV (D) 3 kV
16. An electron is released at rest at point B on the inner surface of the shell. The kinetic energy of electron when it reaches the rod is nearly
 (A) 0.5×10^{-15} J (B) 2.0×10^{-15} J (C) 3.0×10^{-15} J (D) 4.0×10^{-15} J

Space for Rough Work

Paragraph for Questions 17 and 18

A ladder of length ℓ and uniform mass density stands on a frictionless floor and leans against a frictionless wall. It is initially held motionless with its bottom end at an infinitesimal distance from the wall. It is then released where upon the bottom end slides away from the wall and top end slide down the wall, then

17. Angle made by ladder with the wall when it loses contact with the wall is

(A) $\cos^{-1}\left(\frac{2}{3}\right)$ (B) 60° (C) $\sin^{-1}\left(\frac{2}{3}\right)$ (D) 30°

18. Horizontal component of velocity of centre of mass of ladder when it loses contact with the wall is

(A) $2\sqrt{2g\ell}$ (B) $2\sqrt{2g\ell/3}$ (C) $\frac{\sqrt{g\ell}}{3}$ (D) $\sqrt{2g\ell/3}$

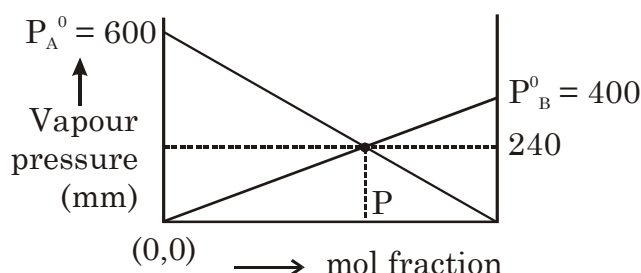
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks : 18)

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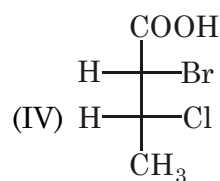
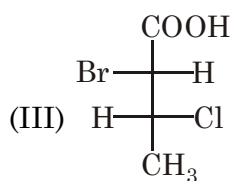
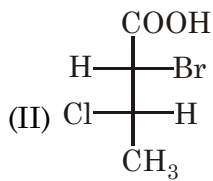
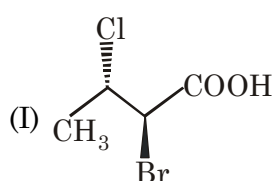
1. Ionic compound AB has (ZnS) Zinc Blend type structure. The edge length of unit cell is 'a'. The correct statement is -
 (A) Nearest distance between cation and anion is $a/2$
 (B) Nearest distance between two cation is $a/\sqrt{2}$
 (C) Nearest distance between two anion is 'a'
 (D) Coordination number of S^{2-} is 8
2. Two volatile liquids A & B form an ideal Binary solution and variation of vapour pressure verses mole fraction in liquid solution is described in the following diagram. Which of the following statement regarding A, B and the solution of A and B is **INCORRECT** -



- (A) Vapour pressure of A in pure state is greater than that of B in pure state at same temperature
- (B) Vapour pressure of solution at P is 480 mm of Hg
- (C) Mol fraction of B in liquid solution at P is 0.6
- (D) Mol fraction of A in vapour phase at equilibrium with solution at 'P' is 0.6.

Space for Rough Work

3. Neither acidic nor basic radical for which compound decolorise acidic KMnO_4 solution.
 (A) KNO_2 (B) Na_2CO_3 (C) FeC_2O_4 (D) BaSO_3
4. 2nd ionisation enthalpy for which 3d transitional element is highest.
 (A) Zn (B) Cu (C) Ni (D) Fe
5. Which of the following amino acid has 2° amine group?
 (A) Asparagine (B) Glutamine (C) Proline (D) Phenyl alanine
6. Identify the pair of compounds having same solubility in H_2O



(A) I and II

(B) I and III

(C) II and IV

(D) I and IV

Space for Rough Work

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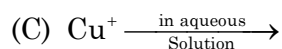
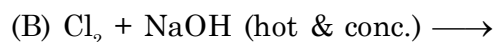
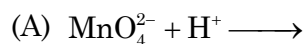
-
7. Which statements is/are correct for ideal gas-
- (A) Final temperature in reversible adiabatic expansion is lesser than in irreversible adiabatic expansion till same final volume
- (B) When heat is supplied to an ideal gas in an isothermal process, kinetic energy of gas molecules will increase
- (C) When an ideal gas is subjected to adiabatic compression it get heated.
- (D) Entropy increases in atomisation of hydrogen gas.
-

Space for Rough Work

8. When negatively charged colloid like As_2S_3 sol is added to positively charged $\text{Fe}(\text{OH})_3$ sol in suitable amounts -
- (A) Both sols are precipitated simultaneously
 - (B) The process is called mutual coagulation
 - (C) They become positively charged colloids
 - (D) They become negatively charged colloids
9. Which of the following sample of reducing agents is/are chemically equivalent to 25ml of 0.2 N KMnO_4 in acidic medium -
- (A) 50 mL of 0.1M FeSO_4 to be oxidized to Fe^{3+}
 - (B) 50 mL of 0.05 M SnCl_2 to be oxidized to Sn^{4+}
 - (C) 25 mL of 0.1M H_3AsO_3 to be oxidized to H_3AsO_4
 - (D) 25 mL of 0.1M H_2O_2 to be oxidized to O_2
10. Which of the following option is **INCORRECT**?
- (A) Among BiI_3 , HgI_2 , PbI_2 , Hg_2I_2 , black colour iodide is BiI_3
 - (B) Among Zn^{2+} , Co^{2+} , Ni^{2+} , Cd^{2+} , the ions that does not dissolve in excess $\text{NH}_{3(\text{aq.})}$ solution is Cd^{2+}
 - (C) BaC_2O_4 on treatment with conc. H_2SO_4 gives both neutral and acidic gas.
 - (D) KNO_3 gives NO_2 gas on heating.

Space for Rough Work

11. Which of the following reaction give products by disproportionation reaction?



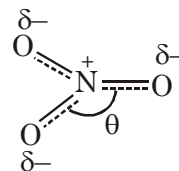
12. Which of the following statement are **CORRECT** ?

(A) 10 volume H_2O_2 solution is equivalent to 6% H_2O_2

(B) Among $\text{H}_4\text{P}_2\text{O}_5$; $\text{H}_4\text{P}_2\text{O}_6$; $\text{H}_4\text{P}_2\text{O}_7$; $(\text{HPO}_3)_n$, total three P–OH linkage is present only in $(\text{HPO}_3)_n$

(C) Among $[\text{AuF}_4]^{-1}$; $[\text{Cu}(\text{CN})_4]^{3-}$; $[\text{Ni}(\text{NH}_3)_6]^{2+}$ & $[\text{Fe}(\text{CN})_5\text{NOS}]^{4-}$, magnetic moment $\sqrt{8}$ B.M. is observed in $[\text{Ni}(\text{NH}_3)_6]^{2+}$ only

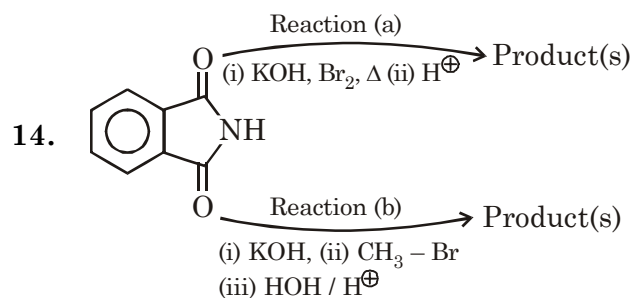
(D) CH_3OCH_3 , NH_3 , H_2O has bond angle less than bond angle θ of



Space for Rough Work

13. Bromine water test is given by?

- (A) Alkene or alkyne (B) Cyclo propane (C) Phenol (D) Aniline



Correct statement about reaction (a) and (b) is / are

- (A) Product(s) of reaction (a) gives isocyanide test
 (B) Product(s) of reaction (b) gives isocyanide test
 (C) Products of reaction (a) gives coupling reaction and also dye test with benzene diazonium chloride
 (D) Product of reaction (b) gives coupling reaction and also dye test with benzene diazonium chloride

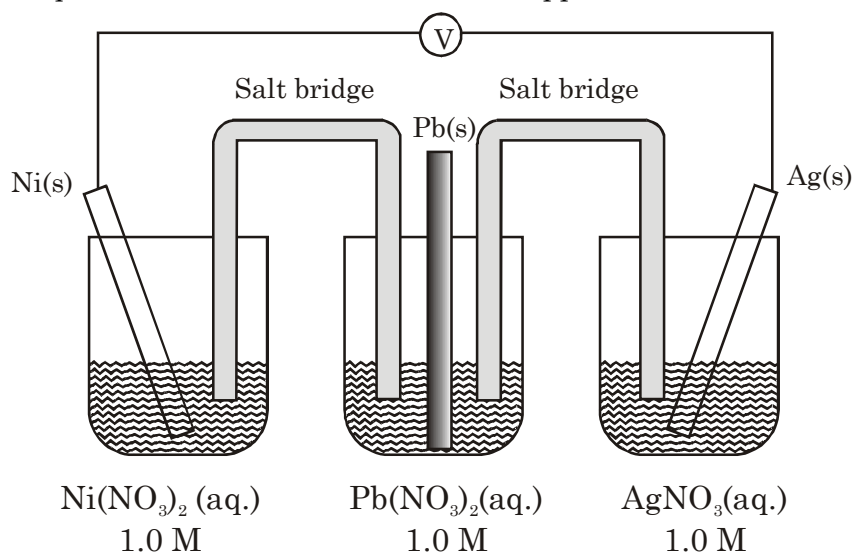
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Paragraph for Questions 15 and 16

The next three questions relate to the illustrated apparatus below :

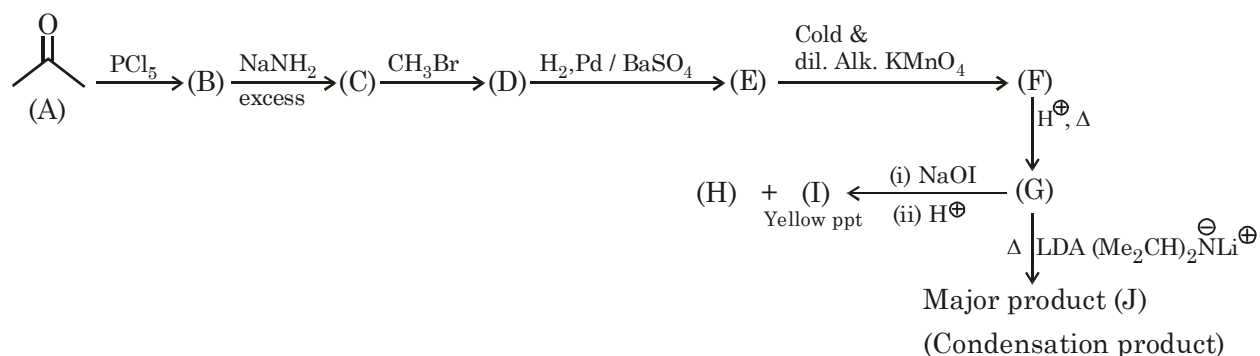


Half cells	E^0 (V at 25°C)	Molar masses
$\text{Ni}^{2+} + 2e^- \rightleftharpoons \text{Ni}$	-0.25V	Ni = 60
$\text{Pb}^{2+} + 2e^- \rightleftharpoons \text{Pb}$	-0.13V	Pb = 207
$\text{Ag}^+ + e^- \rightleftharpoons \text{Ag}$	+0.80	Ag = 108

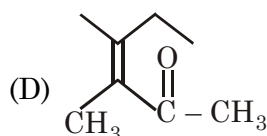
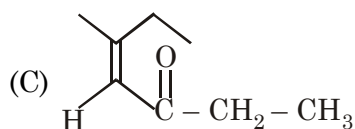
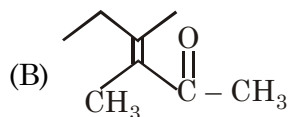
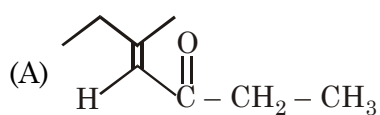
15. When the voltmeter is connected between Ni and Pb electrodes, it reads 0.12 V. After 0.10 mol of electrons have passed through the voltmeter, the mass of Ni electrodes will be-
- (A) 6 grams greater (B) 3 grams greater (C) 6 grams less (D) 3 gram less
16. Based on the information above in which cases would a spontaneous reaction be expected to take place between the metal and the solution paired with it ?
- (I) $\text{Pb(s)} - \text{Ag}^+(\text{aq.})$ (II) $\text{Ag(s)} - \text{Ni}^{2+}(\text{aq.})$ (III) $\text{Pb(s)} - \text{Ni}^{2+}(\text{aq.})$ (IV) $\text{Ag(s)} - \text{Pb}^{2+}(\text{aq.})$
 (V) $\text{Ni(s)} - \text{Pb}^{2+}(\text{aq.})$
- (A) I and II only (B) I and V only (C) II and V only (D) I, III and IV

Space for Rough Work

Paragraph for Questions 17 and 18



17. Major product J is ?



18. Identify the incorrect statement ?

- (A) A and G are homologous
- (B) A and H are functional isomers
- (C) C to D is an example of S_N^2 reaction.
- (D) In B to C formation, E_2 elimination is involved.

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. The number of ways in which the integers from -10 to 10 can be arranged in a sequence such that the absolute value of the numbers in the sequence is non-decreasing is
 (A) 10! (B) 20! (C) $2 \times 10!$ (D) 1024
2. Let α, β and γ be the length of sides of a triangle ΔPQR . The possible value of $\left[\frac{1}{2} + \frac{\alpha}{\beta + \gamma} + \frac{\beta}{\gamma + \alpha} + \frac{\gamma}{\alpha + \beta} \right]$ is equal to (where $[.]$ represent greatest integer function)
 (A) 2 (B) 5 (C) 4 (D) 1
3. If $I_k = \int_{-k\pi}^{k\pi} |\sin x| [\sin x] dx$, $k \in \mathbb{N}$ (where $[.]$ represents greatest integer function and $|\cdot|$ represents absolute value function), then $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{I_k I_{k+1}} =$
 (A) $-\frac{1}{4}$ (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

Space for Rough Work

4. A square matrix of size $n \times n$ is called an orthogonal matrix iff $AA^T = A^TA = I_n$, $n \in \mathbb{N}$.

If $A = \begin{bmatrix} a & b & c \\ c & a & b \\ b & c & a \end{bmatrix}$ is an orthogonal matrix, then magnitude of $(a + b + c)$ is equal to

- (A) 0 (B) 1 (C) 2 (D) 3
5. If the parabola $y = ax^2 + bx + c$ has vertex at $(4, 2)$ and $a \in [2, 4]$ then the ratio of minimum and maximum value of (abc) is equal to-

- (A) $\frac{132}{17}$ (B) $\frac{33}{17}$ (C) $\frac{66}{17}$ (D) $\frac{32}{17}$

6. Let $f(x)$ be a differentiable function such that $f^{2020}(0)$ exists ($f^n(x) = n^{\text{th}}$ order derivative of

$f(x)$ with respect to x). If $f(x)$ satisfies $f(|x| - 1) = \left(\frac{(2019)^x - (2019)^{-x}}{(2019)^x + (2019)^{-x}} \right) f(x)$, $\forall x \in \mathbb{R}$, then $f^{2020}(0)$

is equal to-

- (A) 1 (B) -1 (C) 0 (D) 2020

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
 - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
-
7. Let ABC be a triangle and a, b and c are sides length opposite to vertices A,B and C respectively. If $\vec{\alpha}$ and $\vec{\beta}$ are two non collinear unit vectors satisfying $(a-b)\vec{\alpha} + (b-c)\vec{\beta} + (c-a)(\vec{\alpha} \times \vec{\beta}) = \vec{0}$, then ΔABC is-
- (A) An acute angled triangle
 (B) An obtuse angled triangle
 (C) A right angled triangle
 (D) $R = 2r$ (where R and r are circum radius and in-radius of ΔABC)
8. Which of the following statement is/are true for the curve $(y-2)^2 = 2x^3 - 4$
- (A) $y = 2\sqrt{3}x + 2\sqrt{3} + 2$ is tangent at (1,2)
 (B) $y = 2\sqrt{3}x - 2\sqrt{3} + 2$ is tangent from (1,2)
 (C) No tangent can be drawn from (1,2)
 (D) Exactly two tangents can be drawn from (1,2)

Space for Rough Work

9. Three randomly chosen natural numbers x, y and z satisfy $x + y + z = 10$, then

- (A) probability that z is even is $\frac{1}{2}$ (B) probability that z is even is $\frac{4}{9}$
- (C) probability that z is odd is $\frac{5}{9}$ (D) probability that x is even is $\frac{5}{12}$

10. Let $a \in \mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^7 - 7x + a$, then

- (A) $f(x) = 0$ has 3 real and distinct roots if $a > 6$
- (B) $f(x) = 0$ has only one real root if $a > 6$
- (C) $f(x) = 0$ has 3 real and distinct roots if $a < -6$
- (D) $f(x) = 0$ has 3 real and distinct roots if $-6 < a < 6$

11. The function $y = f(x)$ is the solution of the differential equation

$$\frac{dy}{dx} + \frac{2xy}{x^2 - 1} = \frac{2x}{(x^2 - 1)^2} \text{ in } (1, \infty) \text{ satisfying } f(\sqrt{2}) = 0, \text{ then-}$$

- (A) $f(x)$ is a strictly increasing function in $(1, \infty)$
- (B) $f(x) = \frac{\ln(x^2 - 1)}{(x^2 - 1)}$
- (C) $f(x)$ is a non-monotonic function in $(1, \infty)$
- (D) $f(x)$ is a strictly decreasing function in $(1, \infty)$

Space for Rough Work

12. Let $a_1, a_2, a_3, \dots, a_{100}$ be an arithmetic progression with $a_1 = 5$ and $S_n = \sum_{k=1}^n a_k$. If $\frac{S_m}{S_n}$ is independent of n where $m = 5n$ then possible value of a_2 is/are ($m, n \in \mathbb{N}$) -
 (A) 3 (B) 9 (C) 5 (D) 15
13. Which of the following is/are not the square of a 3×3 matrix with real entries ?
 (A) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -4 \end{bmatrix}$ (B) $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$
14. A variable chord of $\frac{x^2}{9} - \frac{y^2}{b^2} = 1$ subtends 90° at the centre of the hyperbola. If this chord touches a fixed circle concentric with the hyperbola then
 (A) radius of circle is an irrational number for $b = 2$
 (B) radius of circle is a rational number for $b = 5$
 (C) centre of circle is at $(0,0)$
 (D) the circle passes through $\left(\sqrt{3}, \sqrt{\frac{123}{7}}\right)$ for $b = 4$

Space for Rough Work

SECTION-I(iii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

Paragraph for Questions 15 and 16

Let $A(3,4)$, $B(-4,3)$ and $C(4,3)$ be the vertices of $\triangle ABC$. $\triangle ABC$ is inscribed in a circle $S = 0$. Let AD, BE and CF be altitudes from A to BC , B to AC and C to AB respectively. AD, BE and CF are extended to meet $S = 0$ in $D_1(z_1)$, $E_1(z_2)$, $F_1(z_3)$ respectively. (Here z_1, z_2 and z_3 are three complex numbers).

On the basis of above information, answer the following questions :

15. The value of $z_1 z_2 z_3 =$
(A) $75 + 100i$ (B) $-100 + 75i$ (C) $100 + 75i$ (D) $75 - 100i$
16. Orthocenter of $\triangle ABC$ is at
(A) $(2,9)$ (B) $(3,10)$ (C) $(3,11)$ (D) $(9,2)$

Space for Rough Work

Paragraph for Questions 17 and 18

Let A_r represents area of the region bounded between the curve $y^2 = (e^{-kr})x$, $k > 0$, $r \in \mathbb{N}$ and the line $y = mx$, $m \neq 0$, k and m are given constants

On the basis of above information, answer the following questions :

17. $A_1, A_2, A_3, \dots$ are in

- (A) A.P. (B) G.P. (C) H.P. (D) A.G.P.

18. If $\lim_{n \rightarrow \infty} \sum_{r=1}^n A_r = \frac{4}{3(e^{2k} - 1)}$, then the value of m is-

- (A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) 2 (D) 3

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work